

AI/ML INTERNSHIP – DAY 3 SUBMISSION

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Role: AI/ML Intern

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Project Folder: Raritone

1. Objective

The objective of Day 3 was to develop a basic human body detection system and explore how AI can be used for body measurement and fashion-related applications. The task focused on implementing pose detection, understanding measurement logic, and researching AI-based clothing recommendation systems.

2. Task Overview

Task 1 — Human Pose Detection

- Used OpenCV and MediaPipe
- Detected body landmarks
- Displayed skeleton tracking in real time

Task 2 — Body Measurement Logic

Studied how body measurements can be estimated using pose landmarks:

- Shoulder width calculation
- Height estimation
- Waist estimation

Task 3 — AI Research

Explored:

- Virtual try-on AI systems
- Fashion recommendation models

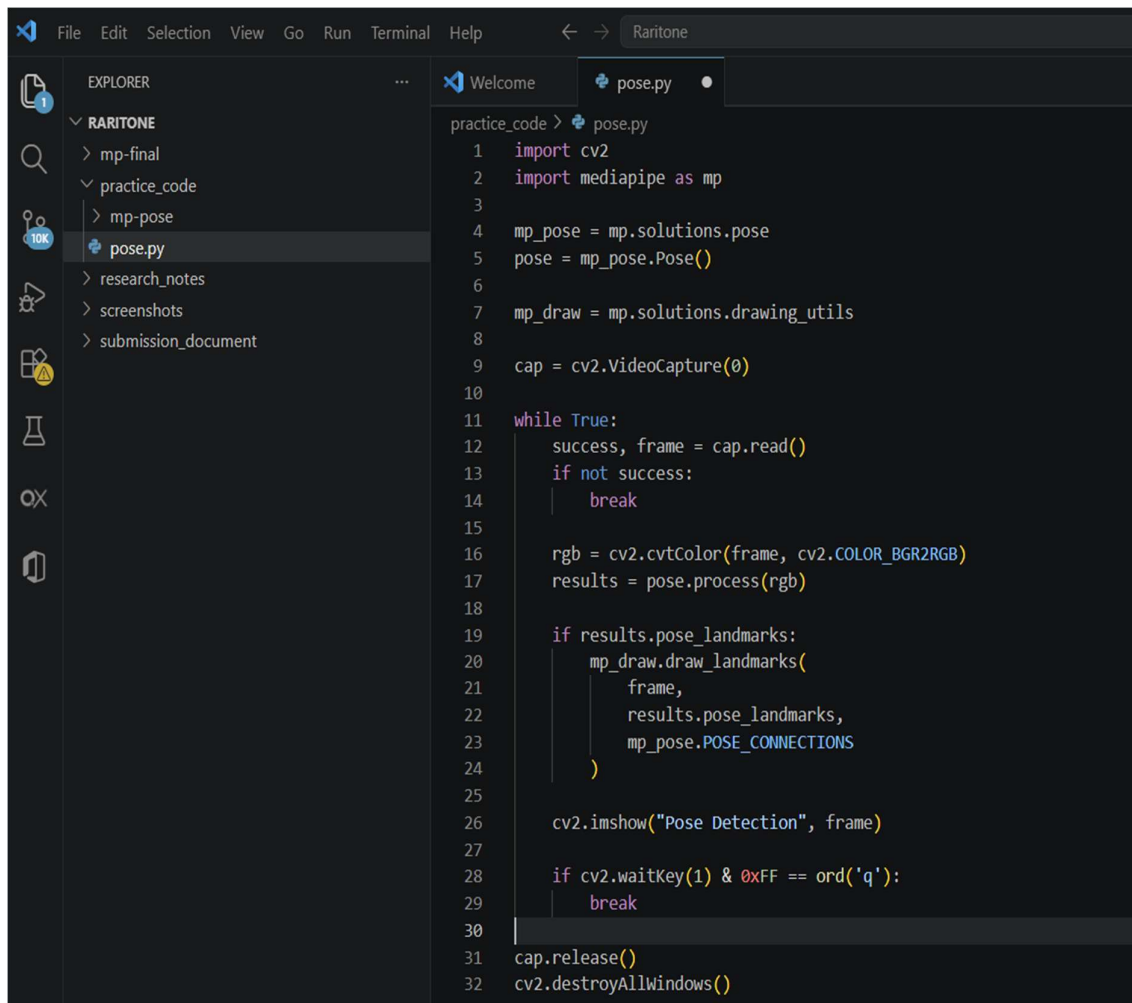
3. Implementation (Task 1)

A real-time human pose detection system was developed using OpenCV and MediaPipe.

Steps Performed

- Captured video using webcam
- Converted frames from BGR to RGB
- Processed frames using MediaPipe Pose model
- Detected body landmarks
- Drew skeletal connections on the body

4. Code Implementation



```
File Edit Selection View Go Run Terminal Help
Raritone

EXPLORER
RARITONE
  > mp-final
  > practice_code
    > mp-pose
      + pose.py
  > research_notes
  > screenshots
  > submission_document

Welcome
pose.py

practice_code > pose.py
1 import cv2
2 import mediapipe as mp
3
4 mp_pose = mp.solutions.pose
5 pose = mp_pose.Pose()
6
7 mp_draw = mp.solutions.drawing_utils
8
9 cap = cv2.VideoCapture(0)
10
11 while True:
12     success, frame = cap.read()
13     if not success:
14         break
15
16     rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
17     results = pose.process(rgb)
18
19     if results.pose_landmarks:
20         mp_draw.draw_landmarks(
21             frame,
22             results.pose_landmarks,
23             mp_pose.POSE_CONNECTIONS
24         )
25
26     cv2.imshow("Pose Detection", frame)
27
28     if cv2.waitKey(1) & 0xFF == ord('q'):
29         break
30
31 cap.release()
32 cv2.destroyAllWindows()
```

5. Output / Deliverable

The system successfully:

- Detected human body landmarks
- Displayed skeleton tracking in real time

